Daily Drilling Report
SPD19B-03
8/22/2014

PETROPARS Oil & Gas Developer

N.I.O.C. Pars Oil & Gas Company

Well Data

Well Name	Operator	Field Name	Well Type 1	Rig	Code
SPD19B-03	Petropars Ltd.	South Pars Gas Field	Development	Deep Driller, DD6	SP19-3B-01-DL-FOR-0...
Report Number	**Hole Section**	**Days LTI (days)**	**Water Depth (m)**	**Original KB Elevation (m)**	**Head Count**
26.0	12 1/4"	171.00	72.83	31.70	96.0
Last Cas	**Daily Cost Total**	**Daily Mud Cost**	**Cum Cost To Date**	**AFE Number**	**Cum Time Log Days (days)**
13 3/8" Shoe, 1,487.00mKB	44,333.81	13,545.60	5,572,197.28		22.85
Depth Start (mKB)	**Depth End (mKB)**	**Depth End (TVD) (mKB)**	**Depth Progress (m)**	**Drilling Hours (hrs)**	**Average ROP (m/hr)**
1,495.00	1,720.00	1,643.57	225.00	15.78	14.3

Operations at Report Time
Drilling @ 1770m.

Operations Summary
RIH directional BHA#8 f/ 1000m to 1495m (Btm) and resume drilling 12 1/4" hole section f/ 1495m to 1720m.

Operations next 24hrs
Cont' drilling 12 1/4" hole section.

Supervisors Contact

Job Contact	Position
Mojtaba Parvaresh	Drilling Supt.
Sajad Tamaddon	Senior Drilling Sup.
Jabbar Mohammadi	Night Drilling Sup.
Hamed Daneshkhah / B.Khalilmoghadam	Drilling Engineer
Ali Sahebnouri	Geologist
R. Kamalpour	POGC Rep.

HSE Data

Type	Last Date	Next Date
Abandon Drill	8/10/2014	9/30/2014
BOP Test	8/20/2014	9/11/2014
Chemical Spill Drill	8/17/2014	10/17/2014
Fire Drill	8/10/2014	9/30/2014
H2S Drill	8/16/2014	9/16/2014
Heli Crash Drill	7/15/2014	9/14/2014
Medical Drill	7/26/2014	9/25/2014
MOB Drill	4/16/2014	6/16/2014
Pit Drill	8/2/2014	9/17/2014
Trip Drill	8/21/2014	10/11/2014

STOP Cards

Number of Reports: 40

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
00:00	01:30	1.50	12 1/4"	D	04	P	RIH directional BHA#8 on 5 7/8" DP f/ 1000m to 1483m. Wash down to 1495m.
01:30	00:00	22....	12 1/4"	D	03	P	Fan bottom. Profile new bit and directional drilling 12 1/4" hole section in rotation & slide mode f/ 1495m to 1720m w/ 750-880 gpm/ 1050-1580 psi/ 5-35 kips/ 40-65 rpm/ 11-22 kft-lbs. Rotation Interval / Slide Interval 1495m-1625m / 1625m-1633m 1633m-1664m / 1664m-1674m 1674m-1712m / 1712m-1720m ** Drilling with NL/NG. ** Pump 20 bbl Hi-Vis pill when Std down. ** B/R and R/D once each Std drilled . ** Take survey before connection.
00:00	00:00		12 1/4"				Aug 23, 2014 00:00 - 06:00 Cont' directional drilling 12 1/4" hole section in rotation & slide mode f/ 1720m to 1770m w/ 750-800 gpm/ 1300-1420 psi/ 7-35 kips/ 45 rpm/ 11-24 kft-lbs. Slide Interval / Rotation Interval 1720m-1722m / 1722m-1742m 1742m-1744m / 1744m-1747m 1747m-1758m / 1758m-1770m >> Last sample @ 1760m: Lst (40 %) + Dol (60 %)

Daily Drilling Report

SPD19B-03

8/22/2014

Mud Information

Depth (mKB)	Type	Density (lb/gal)	Funnel Viscosity (s/qt)	PV Calc (cp)	YP Calc (lbf/100ft²)	Filtrate (mL/30min)
1,720.00	Polymer Mud	9.30	40	15.0	14.000	5.0
pH	**Pf (mL/mL)**	**Mf (mL/mL)**	**Pm (mL/mL)**	**Sand (%)**	**Low Gravity Solids (%)**	**High Gravity Solids (%)**
10.0	0.200	1.000	1.100			
Vis 3rpm	**Vis 6rpm**	**Percent Oil (%)**	**Percent Water (%)**	**Solids (%)**	**Chlorides (mg/L)**	**Filter Cake (/32")**
3	4	0.0	90.0	10.0	24,000.000	1
Gel 10 sec (lbf/100ft²)	**Gel 10 min (lbf/100ft²)**	**Gel 30 min (lbf/100ft²)**	**Oil Water Ratio**	**T (fl) (°C)**	**Active Mud Volume (bbl)**	**Reserve Mud Volume (bbl)**
4.000	6.000	9.000	0/100	55.0	1631.0	2363.0
Cum Mud Added (bbl)	**Mud Lost to Surface (bbl)**	**Mud Lost to Hole (bbl)**	**Avg Background Gas (%)**	**Avg Drill Gas (%)**	**K**	**n**
3550.0	246.0	0.0	0.00		7.480982	0.44

Marine Conditions

Swell Height (m)	Visibility (km)	Wind Direction (°)	Wind Speed (knots)	Barometric Pressure (mmHg)	Temperature - High (°C)
0.50	6.	135	6.0	996.0	34.0

Support Vessels

Vessel Name	Type	Arrival Date	Depart	Note
Marina-1	Supply Boat	8/16/2014 15:15		Stdby Anchorage for DD6 & Sahar 2.
Borkan	Supply Boat	8/22/2014 14:45		Delivere 1 lift Cargo by Snatch. Dep for Sahar 2 @ 1500 Hrs.

Mud Additives

Description	Consumed	Cum Rec	On Loc
Caustic Soda 2014	21.0	244.0	203.0
Defoamer	1.0	8.0	7.0
Citric Acid BG	13.0	150.0	137.0
MI-Cide	5.0	15.0	10.0
Duo-Tec	7.0	115.0	90.0
Soda Ash 25 Kg	70.0	350.0	220.0
Pac-LV	72.0	262.0	125.0
Sodium Bicarbonate	10.0	230.0	220.0
Starch	60.0	680.0	480.0

Bulk Amounts

Supply Item Description	Consumed	Received	Cum On Loc
Potable Water	20.0	17.0	110.0
Drill Water	70.0	0.0	243.0
Barite BB	0.0	0.0	0.0
Barite Bulk	0.0	0.0	249.0
Cement G	0.0	0.0	57.0
Bentonite Bulk	0.0	0.0	8.0
Bentonite BB	0.0	0.0	0.0
Diesel Fuel	8.0	0.0	131.0

Bit/BHA Information

BHA No.	Size (in)	Make	Model	Serial Number	Bit Type	IADC Codes	Nozzles (/32")
8	12 1/4	BEST	M1385	228378	PDC	M432	12/12/12/12/13/13/13/13
TFA (in²)	**Drilling Time (hrs)**	**Bit Revs**	**Depth In (mKB)**	**Depth Out (mKB)**	**Depth Drilled (m)**	**BHA ROP (m/hr)**	**IADC Bit Dull**
0.95	15.78	144,532.20	1,495.00	1,720.00	225.00	14.3	-------
Flow Rate (min) (gpm)	**Flow Rate (max) (gpm)**	**SPP (min) (psi)**	**SPP (max) (psi)**	**WOB (min) (1000lbf)**	**WOB (max) (1000lbf)**	**RPM (min) (rpm)**	**RPM (max) (rpm)**
750	880	1,050.0	1,580.0	9	35	104	179
Circulating Time (hrs)	**Rotating Time (hrs)**	**% Rotating Time (%)**	**% Sliding Time (%)**	**PU HL (max) (1000lbf)**	**Rot HL (max) (1000lbf)**	**SO HL (min) (1000lbf)**	**String Wt (1000lbf)**
21.50	11.90	75.41	24.59	315	280	243	

Drilling Parameters

Depth Start (mKB)	Depth End (mKB)	Drilling Time (hrs)	Drilling Torque	Flow Rate (gpm)	RPM (rpm)	SPP (psi)	WOB (1000lbf)	Drill Str Wt (1000lbf)
1,495.00	1,533.00	3.81	18,000.0	880	65	1,580.0	30	270
1,533.00	1,562.00	2.15	18,000.0	880	60	1,580.0	30	270
1,562.00	1,591.00	1.34	18,000.0	840	60	1,500.0	35	270
1,591.00	1,625.00	2.15	19,000.0	800	60	1,400.0	35	280
1,625.00	1,633.00	1.15		850		1,470.0	10	280
1,633.00	1,664.00	0.70	19,000.0	800	60	1,380.0	30	280
1,664.00	1,674.00	1.60		800		1,350.0	15	280
1,674.00	1,712.00	1.75	22,000.0	750	45	1,050.0	35	189
1,712.00	1,720.00	1.13	22,000.0	800		1,300.0	9	189

Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
3 jts 5 7/8" HWDP	5 7/8	4.250	27.83	226.57
7" Drilling Jar (COR-3019-AD)	7	3.000	5.85	198.74
16 jts 5 7/8" HWDP	5 7/8	4.250	149.23	192.89
X/O	7	2.810	0.79	43.66
1 jt 8 1/4" DC	8 1/4	2.810	9.46	42.87
8 1/2" Float Sub	8 1/2		0.79	33.41
8" NMDC	8	3.500	9.44	32.62

Daily Drilling Report
SPD19B-03
8/22/2014

Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
8" TM - Sub	8		2.03	23.18
8" NMDC	8	3.500	9.46	21.15
12" S. Stabilizer (COR3059AD)	11 7/8	3.125	2.19	11.69
9 5/8" Motor (AKO: 1.00° , Sleeve: 12 1/8")	9 5/8		9.14	9.50
12 1/4" PDC bit	12 1/4		0.36	0.36

Well Control

No.	Depth (mKB)	Stroke (in)	Strokes (spm)	Pres (psi)	Eff (%)	Q (gpm)
2	1,532.00	14.00	30	140.0	97	170
2	1,532.00	14.00	40	190.0	97	227
2	1,532.00	14.00	20	110.0	97	114
3	1,532.00	14.00	20	110.0	97	114
3	1,532.00	14.00	30	140.0	97	170
3	1,532.00	14.00	40	190.0	97	227

Test Date	Last Casing String Run	Test Type	Depth (mKB)	TVD (mKB)	Applied Surface Pressure (psi)
4/8/2014	18 5/8" Shoe, 870.00mKB	Formation Integrity	870.00	843.25	400.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
400.0	8.0	400.00		11.48	1,650.4
Test Date	**Last Casing String Run**	**Test Type**	**Depth (mKB)**	**TVD (mKB)**	**Applied Surface Pressure (psi)**
8/21/2014	13 3/8" Shoe, 1,487.00mKB	Formation Integrity	1,495.00	1,427.34	1,200.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
1,200.0	7.7	1,200.00		14.03	3,413.7

Directional Survey Data

MD (mKB)	Azm (°)	Incl (°)	TVD (mKB)	NS (m)	EW (m)	VS (m)	DLS (°/30m)	Build (°/30m)
1,514.53	246.70	16.20	1,446.09	-180.58	-358.92	401.76	0.21	-0.07
1,543.70	246.50	16.00	1,474.12	-183.79	-366.34	409.84	0.21	-0.21
1,572.73	246.70	15.90	1,502.03	-186.96	-373.66	417.81	0.12	-0.10
1,601.41	246.80	15.70	1,529.63	-190.04	-380.84	425.61	0.21	-0.21
1,630.65	247.10	15.40	1,557.80	-193.11	-388.05	433.44	0.32	-0.31
1,659.53	247.70	16.00	1,585.60	-196.11	-395.26	441.24	0.65	0.62
1,688.53	248.40	16.50	1,613.44	-199.14	-402.79	449.33	0.56	0.52
1,717.53	248.40	17.00	1,641.21	-202.22	-410.56	457.66	0.52	0.52
1,746.33	248.30	17.20	1,668.74	-205.35	-418.43	466.10	0.21	0.21

Geology Information

Formation Name	Prog Top MD (mKB)	Final Top MD (mKB)	Final Top TVD (mKB)
Fahliyan		1,286.50	1,228.82
Hith		1,469.30	1,402.67
Upper Sur.		1,544.50	1,474.91
Upper Dol.		1,616.80	1,544.42
Upper Lst.		1,666.80	1,592.56

General Remarks

Comment

>>> Requirements: (Outstanding of MI MR#12: 2 Pallets of Pac-LV) + (150 MT Cement) + (100 MT FO) + (2 x O-Ring for 10 3/4" Pack off RT) + (200 MT DW).

>>> Sent:
** Marina-1 to Sahar-2: (XO: 2" Weco x 3").

>>> Received:
** Borkan f/ Kish: (7" COR Drilling Jar).